

AN EXACT ANALYSIS FOR VIBRATION OF SIMPLY-SUPPORTED HOMOGENEOUS AND LAMINATED THICK RECTANGULAR PLATES

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A three-dimensional linear, small deformation theory of elasticity solution by the direct method is developed for the free vibration of simply-supported, homogeneous, isotropic, thick rectangular plates. The solution is exact and involves determining a triply infinite sequence of eigenvalues from a doubly infinite set of closed form transcendental equations. As no restrictions are placed on the thickness variation of stresses or displacements, this formulation yields a triply infinite spectrum of frequencies, instead of only one doubly infinite spectrum by thin plate theory and three doubly infinite spectra by Mindlin's thick plate theory. Further, the present analysis yields symmetric thickness modes which neither of the approximate theories can identify. Some numerical results from the two approximate theories are compared with those from the present solution and some important conclusions regarding the effect of the assumptions made in the approximate theories are drawn. The thickness variations of stresses and displacements are also discussed. The analysis is readily extended for laminated plates of isotropic materials. Numerical results are also given for three-ply laminates, and are used to assess the accuracy of thin plate theory predictions for laminates. Extension to general lateral surface conditions and forced vibrations is indicated.

1. INTRODUCTION

Thin plate theory does not take into account either the effect of transverse shear deformation or rotatory inertia, and hence it becomes inaccurate for thicker plates. Mindlin [1] considered both these effects, by assuming that the displacement variation across the thickness is linear for u and v and constant for w . He also had to assign a value to the shearing rigidity factor κ^2 , on suitable physical considerations. His solution does not satisfy the governing differential equations exactly, but does permit the satisfaction of a set of three boundary conditions on each edge. Mindlin, Schacknow and Deresiewicz [2] applied this method to the vibrations of thick rectangular plates with two opposite sides simply supported and the other two edges with various conditions. In this they chose $\kappa^2 = \pi^2/12$, so that Mindlin's theory gives the exact value for the circular frequency of the first antisymmetric mode of thickness shear vibration in an infinite plate. In [2] a simply-supported edge, $x = \text{constant}$, was defined by the conditions

$$w = 0, \quad \frac{\partial v}{\partial z} = 0 \quad \text{and} \quad M_x = 0. \quad (1)$$

The condition $\partial v/\partial z = 0$ along edge $x = \text{constant}$ implies that a line element on that edge, perpendicular to the xy plane does not rotate about the x -axis. This combination of edge conditions, equation (1), amounts to providing an edge support infinitely rigid in its own plane ($w = 0$, $\partial v/\partial z = 0$, along $x = \text{constant}$), but completely flexible to out of plane moments ($M_x = 0$, along $x = \text{constant}$).

In the present paper an exact three-dimensional analysis is developed for the vibration of simply-supported homogeneous and laminated rectangular plates. The boundary conditions for a simply-supported straight edge $x = \text{constant}$ are specified as

$$w = 0, \quad v = 0 \quad \text{and} \quad \sigma_x = 0. \quad (2)$$

These conditions lead to those in equation (1) for approximate theories. For a homogeneous plate, the solution is set up in the form of a double trigonometric series in Cartesian co-ordinates for each of the displacements u, v, w such that the governing differential equation and all edge boundary conditions are identically satisfied. The stress-free homogeneous conditions on the lateral surfaces can also be satisfied identically and this yields closed form transcendental characteristic equations, one for each combination of m and n . Each such equation yields an infinite number of eigenvalues, each one of them representing a different "thickness-mode", the first of these being generally referred to as the "flexural mode" [2].

The analysis of vibrations of laminated plates is a direct extension of that for a homogeneous plate. Each lamina is treated as an individual homogeneous plate and the characteristic equation for the laminate is found by satisfying the homogeneous stress-free conditions on the exterior surfaces, together with equilibrium and continuity conditions on all interfaces. For a p -plied laminate, the order of the characteristic determinant is $6p$. An important feature of this analysis is that all conditions on lateral surfaces, whether interfaces or exterior surfaces, are satisfied identically. A detailed analysis is given for three-ply laminates.

In thin plate theory, only $w(x, y)$ is given freedom, whereas $(\partial u / \partial z)(x, y)$ and $(\partial v / \partial z)(x, y)$ are also given freedom in Mindlin's theory. In the present exact analysis u, v, w at every point in the plate are given freedom. Correspondingly, thin plate theory, Mindlin's theory and the exact solution yield one, three and an infinite number of doubly infinite sets of eigenvalues, respectively. Further, the thin plate and Mindlin analyses do not yield symmetric thickness modes. An exact analysis is necessary to identify the full spectrum of modes and also to assess the approximate theories. The procedure presented in this paper can be applied to study thickness effects in various static and dynamic problems of plates and laminates [3, 4].

Since completion of this work the authors' attention has been drawn to two papers by Woinowsky-Krieger [5] and Mindlin [6]. Woinowsky-Krieger [5] recognized the possibility of using the displacement functions in equation (5) below to study the flexure of homogeneous thick plates under transverse loading. Mindlin [6] also indicated that such functions may be applied for studying waves in homogeneous elastic plates, and obtained the characteristic equation for infinite plates. However, neither reference discusses methods of applying these functions to obtain quantitative data for finite thick plates or to study thickness effects in static and dynamic problems.

2. SOLUTION FOR HOMOGENEOUS PLATES

The basic equations of elasticity in terms of displacements for a body executing simple harmonic oscillations with frequency Ω are

$$\begin{aligned} \nabla^2 u + \frac{1}{1-2\nu} \frac{\partial e}{\partial x} + \frac{d\Omega^2 u}{G} &= 0, \\ \nabla^2 v + \frac{1}{1-2\nu} \frac{\partial e}{\partial y} + \frac{d\Omega^2 v}{G} &= 0, \\ \nabla^2 w + \frac{1}{1-2\nu} \frac{\partial e}{\partial z} + \frac{d\Omega^2 w}{G} &= 0, \end{aligned} \quad (3)$$

where

$$e = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} + \frac{\partial w}{\partial z}.$$

The edge conditions for a simply-supported plate can be specified as

$$\begin{aligned} \text{at } x = 0 \text{ and } a; \quad & \sigma_x = 0, \quad v = 0 \quad \text{and} \quad w = 0; \\ \text{at } y = 0 \text{ and } b; \quad & \sigma_y = 0, \quad u = 0 \quad \text{and} \quad w = 0. \end{aligned} \quad (4)$$

This set of edge conditions is identically satisfied by setting

$$\begin{bmatrix} u \\ v \\ w \end{bmatrix} = h \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \begin{bmatrix} \phi(Z) \cos m\pi X \sin n\pi Y \\ \psi(Z) \sin m\pi X \cos n\pi Y \\ \chi(Z) \sin m\pi X \sin n\pi Y \end{bmatrix}. \quad (5)$$

Substituting equations (5) in (3) one obtains, for each (m, n) combination,

$$\begin{bmatrix} \mathcal{D}^2 + \lambda^2 - g^2 - \frac{M^2}{1-2\nu} & \frac{-MN}{1-2\nu} & \frac{M\mathcal{D}}{1-2\nu} \\ \frac{-MN}{1-2\nu} & \mathcal{D}^2 + \lambda^2 - g^2 - \frac{N^2}{1-2\nu} & \frac{N\mathcal{D}}{1-2\nu} \\ \frac{-M\mathcal{D}}{1-2\nu} & \frac{-N\mathcal{D}}{1-2\nu} & \mathcal{D}^2 + \lambda^2 - g^2 + \frac{\mathcal{D}^2}{1-2\nu} \end{bmatrix} \begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}, \quad (6)$$

where $\mathcal{D}$ denotes the first derivative with respect to Z . The non-trivial solution of equation (6) yields [7] (refer Appendix II)

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} rM & -rM & N & N & M & M \\ rN & -rN & -M & -M & N & N \\ g^2 & g^2 & 0 & 0 & s & -s \end{bmatrix} \begin{bmatrix} Ae^{rZ} \\ Ke^{-rZ} \\ Be^{rZ} \\ Re^{-rZ} \\ Ce^{sZ} \\ Se^{-sZ} \end{bmatrix}, \quad (7)$$

where A, K, B, R, C, S are six arbitrary constants. Using stress-displacement relationships the following expressions for stresses are obtained:

$$\begin{bmatrix} \sigma_x \\ \sigma_y \\ \sigma_z \\ \tau_{xy} \\ \tau_{xz} \\ \tau_{yz} \end{bmatrix} = G \sum_{m=1}^{\infty} \sum_{n=1}^{\infty} \begin{bmatrix} \sigma'_x \sin m\pi X \sin n\pi Y \\ \sigma'_y \sin m\pi X \sin n\pi Y \\ \sigma'_z \sin m\pi X \sin n\pi Y \\ \tau'_{xy} \cos m\pi X \cos n\pi Y \\ \tau'_{xz} \cos m\pi X \sin n\pi Y \\ \tau'_{yz} \sin m\pi X \cos n\pi Y \end{bmatrix}, \quad (8)$$

where

$$\begin{bmatrix} \sigma'_x \\ \sigma'_y \\ \sigma'_z \\ \tau'_{xy} \\ \tau'_{xz} \\ \tau'_{yz} \end{bmatrix} = \begin{bmatrix} -2M^2r & 2M^2r & -2MN & -2MN & \frac{2(s^2-g^2)\nu}{1-2\nu} - 2M^2 & \frac{2(s^2-g^2)\nu}{1-2\nu} - 2M^2 \\ -2N^2r & 2N^2r & 2MN & 2MN & \frac{2(s^2-g^2)\nu}{1-2\nu} - 2N^2 & \frac{2(s^2-g^2)\nu}{1-2\nu} - 2N^2 \\ 2g^2r & -2g^2r & 0 & 0 & (r^2+g^2) & (r^2+g^2) \\ 2MNr & -2MNr & (N^2-M^2) & (N^2-M^2) & 2MN & 2MN \\ (r^2+g^2)M & (r^2+g^2)M & Nr & -Nr & 2Ms & -2Ms \\ (r^2+g^2)N & (r^2+g^2)N & -Mr & Mr & 2Ns & -2Ns \end{bmatrix} \begin{bmatrix} Ae^{rZ} \\ Ke^{-rZ} \\ Be^{rZ} \\ Re^{-rZ} \\ Ce^{sZ} \\ Se^{-sZ} \end{bmatrix}.$$

3. HOMOGENEOUS PLATE WITH STRESS-FREE SURFACES

The lateral surface conditions for free vibrations with stress-free surfaces are

$$\text{at } z = 0 \text{ and } h; \quad \sigma_z = \tau_{xz} = \tau_{yz} = 0. \quad (9)$$

Satisfaction of conditions (9) yields a set of six homogeneous equations for each (m, n) . These can be put in the form

$$\begin{bmatrix} A \\ K \\ B \\ R \\ C \\ S \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \quad (10)$$

where

$$[L(Z, \beta)] = \beta \begin{bmatrix} 2g^2 r & -2g^2 r & 0 & 0 & (r^2 + g^2) & (r^2 + g^2) \\ (r^2 + g^2)M & (r^2 + g^2)M & Nr & -Nr & 2Ms & -2Ms \\ (r^2 + g^2)N & (r^2 + g^2)N & -Mr & Mr & 2Ns & -2Ns \end{bmatrix} [f], \quad (11)$$

and

$$[f] = [e^{rz}, e^{-rz}, e^{rz}, e^{-rz}, e^{sz}, e^{-sz}] \text{ is a diagonal matrix.}$$

For a non-trivial solution of the problem the determinant of $\{[L(0, 1)], [L(1, 1)]\}$ must be zero and this yields the characteristic equation,

$$[8g^2 rs(r^2 + g^2)^2(1 - \cosh r \cosh s) + \{16g^4 r^2 s^2 + (r^2 + g^2)^4\} \sinh r \sinh s] \sinh r = 0. \quad (12)$$

Solution of this transcendental equation for each combination of m and n yields an infinite sequence of eigenvalues. As in thin plate and Mindlin theories, and also in the exact theory, the eigenvalues depend only on ν and g , but not on individual values of m , n , a/b or h/a . Some of the eigenvalues may lead to either pure distortional modes in which the dilatation $\partial u/\partial x + \partial v/\partial y + \partial w/\partial z = 0$ or to pure dilatational modes in which all the rotations $(\partial u/\partial y - \partial v/\partial x)$, $(\partial u/\partial z - \partial w/\partial x)$, $(\partial v/\partial z - \partial w/\partial y)$ vanish. It is possible to identify such modes, if they exist, by a direct study of equation (7) along with the appropriate surface conditions.

3.1. DISTORTIONAL MODES

From equation (7) it is readily seen that, presuming $\nu \neq 0.5$, for all distortional modes $C = S = 0$. $\lambda = \sqrt{g^2 + \alpha^2 \pi^2}$ ($\alpha = 0, 1, 2, \dots$), arising from $r = i\alpha\pi$, are obvious roots of the characteristic equation (12). Correspondingly,

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = A \begin{bmatrix} M \sin \alpha\pi Z \\ N \sin \alpha\pi Z \\ -\alpha\pi \cos \alpha\pi Z \end{bmatrix} + B \begin{bmatrix} N \cos \alpha\pi Z \\ -M \cos \alpha\pi Z \\ 0 \end{bmatrix}, \quad (13)$$

where $A \neq 0$ only when $r^2 + g^2 = 0$. The modes with $A = 0$ correspond to thickness-twist modes in which displacement $w = 0$. Mindlin's thickness-twist mode corresponds to the first exact antisymmetric thickness-twist mode ($\alpha = 1$) and the relevant eigenvalue is also exact, if the constant κ^2 is taken as $\pi^2/12$. Whenever the parameter g/π is an integer, a distortion mode with non-zero w can exist with any combination of A and B . For $\nu = 0.5$, i.e. for incompressible materials, all possible modes are obviously distortional.

3.2. DILATATIONAL MODES

Equation (7) again yields the condition $A = K = B = R = 0$ for the possibility of pure dilatational modes. Further examination shows that no root of equation (12) can yield a dilatational mode for stress-free conditions on the surfaces except when the Poisson's ratio $\nu = 0$. In this case $\lambda = g/\sqrt{2}$, arising from $s = 0$, and

$$\{\phi, \psi, \chi\} = C\{M, N, 0\}. \quad (14)$$

It is also of interest to note that $s = 0$ leads to dilatational waves in infinite bodies irrespective of the value of ν .

3.3. VERY SHORT WAVES

For very short waves, i.e. as $g \rightarrow \infty$, the characteristic equation can be simplified to

$$\left(2 - \frac{c^2}{c_s^2}\right)^4 - 16\left(1 - \frac{c^2}{c_s^2}\right)\left\{1 - \left(\frac{1-2\nu}{2-2\nu}\right)\frac{c^2}{c_s^2}\right\} = 0, \quad (15)$$

where $1/c^2 = 1/c_x^2 + 1/c_y^2$,

$$c_s = \sqrt{G/d}, \quad c_x = \Omega m \pi / a, \quad c_y = \Omega n \pi / b.$$

Here a/m , b/n are wavelengths in x , y directions and c_x , c_y are wave velocities in the x , y directions and c_s is the velocity of distortion waves in an infinite body. Equation (15) is identical with the characteristic equation for Rayleigh's plane surface waves [9]. Thus c is equal to the velocity of plane surface waves; it varies from $0.9194c_s$ for $\nu = 0.25$, to $0.9553c_s$ for $\nu = 0.5$. c is known to be independent of plate thickness and wavelength.

3.4. OTHER MODES

For all modes other than pure distortional modes,

$$\begin{bmatrix} A \\ K \\ B \\ R \\ C \\ S \end{bmatrix} = T \begin{bmatrix} -(r^2 + g^2)^3 \sinh s + 4g^2 rs(r^2 + g^2)(\cosh s - e^{-r}) \\ (r^2 + g^2)^3 \sinh s + 4g^2 rs(r^2 + g^2)(\cosh s - e^r) \\ 0 \\ 0 \\ -8g^4 r^2 s \sinh r + 2g^2 r(r^2 + g^2)^2 (\cosh r - e^{-s}) \\ -8g^4 r^2 s \sinh r - 2g^2 r(r^2 + g^2)^2 (\cosh r - e^s) \end{bmatrix}, \quad (16)$$

where T is an arbitrary constant.

3.5. NUMERICAL RESULTS

The transcendental equation (12) was solved by the Newton-Raphson method. Table 1 gives the eigenvalues obtained by classical thin plate theory, Mindlin's theory [2] and the present exact solution. All the three eigenvalues by Mindlin's analysis and only the first nine exact eigenvalues are tabulated. Table 2 gives the distributions of stresses and displacements across thickness for a plate with $mh/a = nh/b = 0.3$ and $\nu = 0.3$. Table 3 gives the relative magnitudes of stresses $\sigma_{x, \max}$, $\sigma_{z, \max}$ and $\tau_{xz, \max}$ and displacements $u_{\max}$ and $w_{\max}$. From these data the following conclusions can be drawn.

(i) It is confirmed that thin plate theory overestimates the frequencies which it identifies. The errors of thin plate theory increase with increasing " g ".

(ii) The first (flexural) eigenvalues by Mindlin's theory [2] are only slightly lower than the first exact eigenvalues. This could be due to the choice of $\kappa^2 (= \pi^2/12)$. With slightly higher values of κ^2 (about 0.88) Mindlin's flexural eigenvalues can be made equal to the exact values.

TABLE 1

Eigenvalues (λ) for homogeneous plates

ν	$\frac{g^2}{\pi^2}$	Thin plate theory I-A	Mindlin's theory [2]				Present exact analysis							% error in thin plate values§	
			Mindlin's theory [2]				Present exact analysis								
			I-A	II-A	III-A	I-A†	I-S†	II-S	II-A‡	III-A	III-S	IV-S†	V-S		IV-A
0.3	0.0125	0.06020	0.05887	3.1494	3.2124	0.05893	0.35124	0.59314	3.1494	3.2079	5.7980	6.2930	6.3930	9.4232	2.15
0.3	0.02	0.09632	0.09300	3.1729	3.2538	0.09315	0.44429	0.74983	3.1729	3.2465	5.7632	6.2989	6.4461	9.4223	3.40
0.3	0.05	0.24079	0.22176	3.2192	3.4112	0.22260	0.70248	1.1827	3.2192	3.3933	5.6652	6.3223	6.6178	9.4193	8.17
0.3	0.08	0.38527	0.34018	3.2648	3.5580	0.34207	0.88858	1.4943	3.2648	3.5298	5.6010	6.3457	6.7563	9.4174	12.6
0.3	0.1	0.48159	0.41440	3.2949	3.6510	0.41714	0.99346	1.6654	3.2949	3.6160	5.5688	6.3612	6.8384	9.4166	15.5
0.3	0.13	0.62606	0.51974	3.3396	3.7842	0.52391	1.1327	1.8936	3.3396	3.7393	5.5315	6.3845	6.9512	9.4162	19.5
0.3	0.18	0.86686	0.68208	3.4126	3.9926	0.68893	1.3329	2.2171	3.4126	3.9310	5.4903	6.4230	7.1199	9.4174	25.8
0.3	0.2	0.96317	0.74312	3.4414	4.0720	0.75111	1.4050	2.3320	3.4414	4.0037	5.4795	6.4383	7.1825	9.4184	28.2
0.3	0.26	1.2521	0.91520	3.5264	4.2982	0.92678	1.6019	2.6407	3.5264	4.2099	5.4621	6.4842	7.3574	9.4236	35.1
0.3	0.32	1.5411	1.0735	3.6094	4.5098	1.0889	1.7772	2.9066	3.6094	4.4013	5.4635	6.5297	7.5179	9.4315	41.5
0.3	0.50	2.4079	1.4890	3.8476	5.0804	1.5158	2.2214	3.5306	3.8476	4.9086	5.5554	6.6643	7.9416	9.4698	58.9
0.2	0.02	0.09010	0.08728	3.1729	3.2432	0.08734	0.44429	0.70211	3.1729	3.2386	5.0926	6.2989	6.3620	9.3960	3.12
0.4	0.02	0.10403	0.10001	3.1729	3.2680	0.10025	0.44429	0.80816	3.1729	3.2566	6.2482	6.2989	7.7994	9.4327	3.77

† A and S denote modes which are antisymmetric and symmetric about mid-plane, respectively.

‡ Thickness-twist modes.

§ % error = $100 \times \left\{ \frac{\lambda \text{ by thin plate theory}}{\text{first exact } \lambda} - 1 \right\}$.

TABLE 2

Variations of σ_x , σ_z , τ_{xy} , τ_{xz} , u and w across the thickness for homogeneous plate

Z	$\sigma_x/\sigma_x(0)$	$\sigma_z/\sigma_z(0.4)$	$\tau_{xy}/\tau_{xy}(0)$	$\tau_{xz}/\tau_{xz}(0.5)$	$u/u(0)$	$w/w(0)$
Thin plate theory and the 3 thickness modes of Mindlin's theory						
0	1.0	—	1.0†	0	1.0	1.0†
0.1	0.8	—	0.8	0.36	0.8	1.0
0.2	0.6	—	0.6	0.64	0.6	1.0
0.3	0.4	—	0.4	0.84	0.4	1.0
0.4	0.2	—	0.2	0.96	0.2	1.0
0.5	0	—	0	1.0	0	1.0
Exact theory						
First antisymmetric thickness-mode (I-A)						
0	1.0	0	1.0	0	1.0	1.0
0.1	0.7676	1.5336	0.7561	0.3750	0.7561	1.0249
0.2	0.5571	2.0238	0.5420	0.6549	0.5420	1.0426
0.3	0.3627	1.7578	0.3496	0.8426	0.3496	1.0543
0.4	0.1787	1.0	0.1713	0.9625	0.1713	1.0610
0.5	0	0	0	1.0	0	1.0631
Second antisymmetric thickness-mode (II-A)						
0	1.0	—	—	0	1.0	—
0.1	0.9511	—	—	0.3090	0.9511	—
0.2	0.8090	—	—	0.5878	0.8090	—
0.3	0.5878	—	—	0.8090	0.5878	—
0.4	0.3090	—	—	0.9511	0.3090	—
0.5	0	—	—	1.0	0	—
Third antisymmetric thickness-mode (III-A)						
0	1.0	0	1.0	0	1.0	1.0
0.1	0.9174	1.3202	0.9892	0.2925	0.9892	0.8148
0.2	0.7638	1.8691	0.8655	0.5703	0.8655	0.6166
0.3	0.5481	1.7063	0.6409	0.7984	0.6409	0.4426
0.4	0.2863	1.0	0.3407	0.9479	0.3407	0.3243
0.5	0	0	0	1.0	0	0.2825
Second symmetric thickness-mode (II-S)						
0	1.0	0	1.0	0‡	1.0	1.0
0.1	1.0161	0.3611	1.0438	0.7425‡	1.0438	0.8280
0.2	1.0297	0.6534	1.0797	1.0‡	1.0797	0.6377
0.3	1.0400	0.8684	1.1065	0.8813‡	1.1065	0.4333
0.4	1.0464	1.0	1.1230	0.5057‡	1.1230	0.2191
0.5	1.0486	1.0443	1.1286	0‡	1.1286	0

$$\nu = 0.3, mh/a = nh/b = 0.3, g^2 = 0.18 \pi^2.$$

† Note that for thickness-twist mode (II-A), $\tau_{xy} = 0$ (when $M = N$) and $w = 0$ throughout.

‡ $\tau_{xz}/\tau_{xz}(0.2)$ and not $\tau_{xz}/\tau_{xz}(0.5)$.

(iii) The thickness-twist eigenvalues (II-A) obtained by Mindlin's theory (with $\kappa^2 = \pi^2/12$) coincide with the corresponding exact values. The thickness-shear (III-A) eigenvalues by Mindlin's theory are only slightly higher than the third exact antisymmetric eigenvalues.

(iv) Displacement distributions across thickness assumed in thin plate and Mindlin theories are significantly different from the true distributions. For the flexural modes the deviations increase with g .

- (v) Ratios of successive eigenvalues, $\lambda_{i+1}(g)/\lambda_i(g)$, decrease as the relevant "g" is increased.
- (vi) Complexity of the stress and displacement distributions across thickness increases for higher-order thickness modes.
- (vii) As the order of thickness modes increases, the relative magnitudes of the transverse stresses $\sigma_{z, \max}$, $\tau_{xz, \max}$ and $\tau_{yz, \max}$ compared to the inplane stresses $\sigma_{x, \max}$ and $\sigma_{y, \max}$ show considerable increase and in fact the transverse stresses do overtake the inplane stresses at some stage.
- (viii) Poisson's ratio effects on errors in thin plate theory are small.

TABLE 3

Relative magnitudes of stresses and displacements in antisymmetric thickness modes for homogeneous plates

	$\frac{\sigma_{z, \max}}{\sigma_{x, \max}}$	$\frac{\tau_{xz, \max}}{\sigma_{x, \max}}$	$\frac{u_{\max}}{w_{\max}}$
Thin plate theory	0	0.3625	0.4712
Mindlin's theory:			
flexural, I-A	0	0.328	0.3212
thickness-twist, II-A	0	1.309	1/0
thickness-shear, III-A	0	0.777	4.63
Exact analysis:			
thickness mode I-A	0.0363	0.3084	0.3389
thickness mode II-A	0	1.6667	1/0
(thickness-twist)			
thickness mode III-A	0.245	1.0302	2.0503

$$\nu = 0.3, mh/a = nh/b = 0.3, g^2 = 0.18\pi^2.$$

4. LAMINATED PLATES WITH FREE SURFACES

As mentioned earlier, extension of the analysis to laminated plates is formally straightforward. Equations (3), (5) and (6) are valid for individual plies, the quantities for the j th ply being indicated by adding the subscript "j". In a p plied laminate, the lateral surface conditions to be satisfied are

$$\left. \begin{aligned} &\text{at } Z = 0; \quad \sigma_{z_1} = \tau_{xz_1} = \tau_{yz_1} = 0; \\ &\text{at } Z = H_p; \quad \sigma_{z_p} = \tau_{xz_p} = \tau_{yz_p} = 0; \\ &\text{and, at interfaces } Z = H_j \ (j = 1, 2, \dots, p-1), \\ &\text{equilibrium conditions } \sigma_{z_j} - \sigma_{z_{j+1}} = \tau_{xz_j} - \tau_{xz_{j+1}} = \tau_{yz_j} - \tau_{yz_{j+1}} = 0 \\ &\text{and continuity conditions } u_j - u_{j+1} = v_j - v_{j+1} = w_j - w_{j+1} = 0. \end{aligned} \right\} \quad (17)$$

Satisfying the above conditions, equation (17), a characteristic determinantal equation of the order $6p$ is obtained. For a three-ply laminate this can be put in the form

$$\begin{vmatrix} [L(0, 1)]_1 & [0] & [0] \\ [L(H_1, \bar{G}_1)]_1 & [L(H_1, -\bar{G}_2)]_2 & [0] \\ [J(H_1, 1)]_1 & [J(H_1, -1)]_2 & [0] \\ [0] & [L(H_2, -\bar{G}_2)]_2 & [L(H_2, \bar{G}_3)]_3 \\ [0] & [J(H_2, -1)]_2 & [J(H_2, 1)]_3 \\ [0] & [0] & [L(H_3, 1)]_3 \end{vmatrix} = 0, \quad (18)$$

where

$[0]$ is a (3×6) null matrix and

$$[J(Z, \beta)]_j = \beta \begin{bmatrix} rM & -rM & N & N & M & M \\ rN & -rN & -M & -M & N & N \\ g^2 & g^2 & 0 & 0 & s & -s \end{bmatrix}_j [f]_j.$$

Extension of equation (18) to multi-ply laminates is straightforward.

4.1. SOME NUMERICAL RESULTS

Numerical results are presented in Table 4 for three-ply laminates symmetric about middle planes with $g_1^2 = g_3^2 = 0.0002\pi^2$ and $g_2^2 = 0.0128\pi^2$. To study the effect of differences in moduli, laminates with modular ratios $G_1/G_2 = G_3/G_2 = 1, 2, 5, 15$, $d_1 = d_2 = d_3$ and $\nu_1 = \nu_2 = \nu_3 = 0.3$ are considered. To study the effect of differences in densities, laminates with $d_1/d_2 = d_3/d_2 = 1, 2, 3$, $G_1/G_2 = G_3/G_2 = 15$ and $\nu_1 = \nu_2 = \nu_3 = 0.3$ are considered. A laminate with $\nu_1 = \nu_3 = 0.2$ and $\nu_2 = 0.3$, $G_1 = G_2 = G_3$ and $d_1 = d_2 = d_3$ is considered to examine the effect of differences in Poisson's ratio.

From the data in Table 4, one may make the following observation in addition to the conclusions drawn for homogeneous plates: errors in eigenvalues due to thin plate approximations are significantly influenced by the modular ratios, but not by the values of the Poisson's ratios or in case of small M and N by density ratios. The thin plate errors increase with increasing moduli of outer plies.

TABLE 4

Eigenvalues (λ) for three-ply laminates obtained by thin plate theory and present exact solution

ν_1	$\frac{d_1}{d_2}$	$\frac{G_1}{G_2}$	thin plate theory [8]	first exact eigenvalue	% error†
0.3	1	1	0.077054	0.07452	3.4
0.3	1	2	0.093994	0.089986	4.45
0.3	1	5	0.13239	0.123072	7.57
0.3	1	15	0.215642	0.183664	17.41
0.3	2	15	0.196852	0.167574	17.47
0.3	3	15	0.18225	0.155082	17.52
0.2	1	1	0.074666	0.072298	3.28

$$d_1 = d_3, G_1 = G_3, \nu_1 = \nu_3, \nu_2 = 0.3, g_1^2 = g_3^2 = 0.0002\pi^2, g_2^2 = 0.0128\pi^2, \lambda = \Omega \sqrt{d_2 h_2^3 / G_2}.$$

$$\dagger \% \text{ error} = 100 \times \left(\frac{\text{thin plate theory value}}{\text{first exact eigenvalue}} - 1 \right).$$

5. HOMOGENEOUS PLATES WITH OTHER SURFACE CONDITIONS

The present formulation is such that any lateral surface condition which can be well represented by double Fourier series can be satisfied. In this section three simple cases are discussed. Even more general cases of elastic foundations or mass distributions on the surfaces present no difficulty.

5.1. PLATE HELD BETWEEN TWO SMOOTH RIGID SURFACES

The lateral surface conditions in this case are

$$\text{on } Z = 0, 1: \quad w = 0, \quad \tau_{xz} = \tau_{yz} = 0. \quad (19)$$

Their satisfaction yields the characteristic equation

$$\sinh s \sinh^2 r = 0. \quad (20)$$

The eigenvalues $\lambda = \sqrt{g^2 + \alpha^2 \pi^2}$ ($\alpha = 0, 1, 2, \dots$) arising from $r = i\alpha\pi$, correspond to pure distortional modes with

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} \alpha\pi M \cos \alpha\pi Z \\ \alpha\pi N \cos \alpha\pi Z \\ g^2 \sin \alpha\pi Z \end{bmatrix} A + \begin{bmatrix} N \cos \alpha\pi Z \\ -M \cos \alpha\pi Z \\ 0 \end{bmatrix} B. \quad (21)$$

When $A = 0$, the modes are thickness-twist modes and are identical with those obtained earlier for stress-free surfaces.

Another set of eigenvalues, $\lambda = \sqrt{[(2-2\nu)/(1-2\nu)](g^2 + \alpha^2 \pi^2)}$ ($\alpha = 0, 1, 2, \dots$), arising from $s = i\alpha\pi$ correspond to pure dilatational modes with

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} M \cos \alpha\pi Z \\ N \cos \alpha\pi Z \\ -\alpha\pi \sin \alpha\pi Z \end{bmatrix} C. \quad (22)$$

5.2. PLATE BONDED TO RIGID SURFACES

The lateral surface conditions are

$$\text{on } Z = 0, 1: \quad u = v = w = 0. \quad (23)$$

This yields the characteristic equation

$$\{(r^2 s^2 + g^4) \sinh r \sinh s + 2g^2 rs(1 - \cosh r \cosh s)\} \sinh r = 0. \quad (24)$$

$\lambda = \sqrt{g^2 + \alpha^2 \pi^2}$ ($\alpha = 1, 2, 3, \dots$) are eigenvalues for pure distortional modes of the thickness-twist type with

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} N \sin \alpha\pi Z \\ -M \sin \alpha\pi Z \\ 0 \end{bmatrix} B. \quad (25)$$

In this case, no pure dilatational modes occur.

5.3. PLATE ON ELASTIC FOUNDATION

If the plate is supported at $Z = 0$ on a smooth elastic foundation, with normal spring constant k , the lateral surface conditions are

$$\begin{aligned} \text{on } Z = 0: \quad & \sigma_z - kw = 0, \quad \tau_{xz} = \tau_{yz} = 0; \\ \text{on } Z = 1: \quad & \sigma_z = \tau_{xz} = \tau_{yz} = 0. \end{aligned} \quad (26)$$

The characteristic equation in this case turns out to be

$$\begin{aligned} & ([8g^2 rs(r^2 + g^2)^2(1 - \cosh r \cosh s) + \{16g^4 r^2 s^2 + (r^2 + g^2)^4\} \sinh r \sinh s] + \\ & + [ks(g^2 - r^2)\{4g^2 rs \cosh r \sinh s - (r^2 + g^2)^2 \cosh s \sinh r\}]) \sinh r = 0. \end{aligned}$$

It is interesting to note that in spite of the elastic foundation, the pure thickness-twist modes with zero w and $\lambda = \sqrt{g^2 + \alpha^2 \pi^2}$ ($\alpha = 0, 1, 2, \dots$) persist.

6. FORCED VIBRATIONS

When the plate (or laminate) is acted upon by harmonic forcing functions distributed in any manner on the lateral surfaces, particular solutions can be obtained. The forcing functions are expressed in double Fourier series in (X, Y) and incorporated as appropriate boundary conditions on the lateral surfaces, so that the right-hand sides in equations analogous to

equation (10) are non-zero. The solution of the resulting simultaneous equations yields the displacements, etc., for each (m, n) . Summation of the solutions with respect to (m, n) completes the analysis.

7. CONCLUDING REMARKS

An exact closed form characteristic equation has been derived for obtaining natural frequencies of thick, simply-supported, homogeneous or laminated rectangular plates. In the description of a simple support, a condition $v = 0$ (along a straight edge $x = \text{constant}$) was chosen in preference to an alternative description, $\tau_{xy} = 0$. This resulted in a closed form characteristic equation. Because of no restriction being placed on the displacement pattern across thickness, the exact solution yielded an infinite number of doubly infinite spectra of natural frequencies, as against the three doubly infinite spectra by Mindlin's analysis and one doubly infinite spectrum by thin plate theory.

For flexural, thickness-twist and thickness-shear frequencies, Mindlin's theory is accurate even at higher values of g , i.e. even for larger thicknesses and higher modes, while the thin plate theory flexural eigenvalues are reasonably accurate (say to within 1%) only for small values of " g " not exceeding 1/5. Essentially, this means that accepting 1% as satisfactory accuracy, $g = 1/5$ sets the limit in terms of a plate parameter combining thickness and mode order, for the applicability of thin plate approximations. However, the stress and displacement distributions across thickness in both thin plate and Mindlin theories are much different from the true distributions.

In laminated plates the relative modular ratios of the plies have significant effect on the accuracy of thin plate theory estimates.

Poisson's ratio seems to have relatively small effect on the errors in thin plate eigenvalues for homogeneous or laminated plates.

Finally, a conclusion of practical importance. Mindlin's approximation and simple theory are fully justified if one is seeking only the flexural, thickness-twist and thickness-shear frequencies, corresponding to each (m, n) . The exact three-dimensional analysis is necessary if the full spectrum of modes needs to be established.

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REFERENCES

1. R. D. MINDLIN 1951 *J. appl. Mech.* **18**, 31. Influence of rotatory inertia and shear on flexural motions of isotropic elastic plates.
2. R. D. MINDLIN, A. SCHACKNOW and H. DERESIEWICZ 1956 *J. appl. Mech.* **23**, 431. Flexural vibrations of rectangular plates.
3. S. SRINIVAS, A. K. RAO and C. V. JOGA RAO 1969 *Z. angew. Math. Mech.* **49**, 449. Flexure of simply supported thick homogeneous and laminated rectangular plates.
4. S. SRINIVAS and A. K. RAO 1969 *AIAA J.* **7**, 1645. Buckling of thick rectangular plates.

5. S. WOINOWSKY-KRIEGER 1933 *Ingénieur-Archiv*, **4**, 203. Der spannungszustand in dicken elastischen platten.
6. R. D. MINDLIN 1958 *Proc. Symp. nav. struct. Mech.* **1**, 199. Waves and vibrations in isotropic, elastic plates.
7. W. E. BOYCE and R. C. DIPRIMA 1965 *Elementary Differential Equations and Boundary Value Problems*. Chapter 7. New York: John Wiley and Sons.
8. K. S. PISTER 1959 *J. acoust. Soc. Am.* **31**, 233. Flexural vibrations of thin laminated plates.
9. S. TIMOSHENKO and J. N. GOODIER 1951 *Theory of Elasticity*. Chapter 15. New York: McGraw-Hill.

APPENDIX I

NOMENCLATURE

a, b, h	length, width, thickness
d	mass density
G	shear modulus
g	$\sqrt{M^2 + N^2}$
M, N	$m\pi h/a, n\pi h/b$
r	$\sqrt{g^2 - \lambda^2}$
s	$\sqrt{g^2 - \lambda^2(1 - 2\nu)/(2 - 2\nu)}$
u, v, w	displacements in x, y, z directions
X, Y, Z	$x/a, y/b, z/h$
λ	eigenvalue = $\Omega\sqrt{dh^2/G}$
ν	Poisson's ratio
Ω	angular frequency

FOR LAMINATED PLATES

d_j, G_j, h_j, ν_j	mass density, shear modulus, thickness and Poisson's ratio of j th ply
d, G, h	some (could be that of one of the plies) mass density, shear modulus and thickness used to non-dimensionalize d_j, G_j and h_j
$\bar{d}_j, \bar{G}_j, \bar{h}_j$	$d_j/d, G_j/G, h_j/h$
H_j	$\left(\sum_{k=1}^j h_k \right) / h$
r_j	$\sqrt{g^2 - \lambda^2 \bar{d}_j / \bar{G}_j}$
s_j	$\sqrt{g^2 - \lambda^2(1 - 2\nu_j) \bar{d}_j / \{(2 - 2\nu_j) \bar{G}_j\}}$

Figure 1 explains the co-ordinate system.

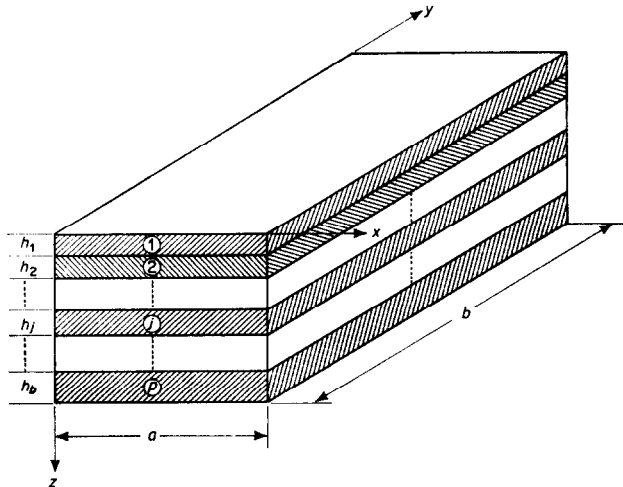


Figure 1. Co-ordinate system.

APPENDIX II

SOLUTION OF EQUATION (6)

For non-trivial solution of equation (6) the (3×3) determinant on the left-hand side of it is zero. This yields for $\mathcal{D}$ two double roots $\pm r$ ($r = \sqrt{g^2 - \lambda^2}$) and two single roots $\pm s$ ($s = \sqrt{g^2 - \lambda^2(1 - 2\nu)/(2 - 2\nu)}$). To find the eigenvectors $\{\phi, \psi, \chi\}$ corresponding to the root $+r$, it is assumed that

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} e^{rz}, \quad (\text{A1})$$

where P_1, P_2, P_3 are yet unknown constants. Substitution of equation (A1) in equation (6) yields, after some manipulation,

$$\begin{bmatrix} M & N & -r \\ M & N & -r \\ M & N & -r \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (\text{A2})$$

The rank of the (3×3) matrix on the left-hand side of equation (A2) is one and therefore the solution of form (A1) can yield two eigenvectors each with an arbitrary constant, which are necessary and sufficient for a double root. The only relation between the constants P_1, P_2, P_3 is

$$MP_1 + NP_2 - rP_3 = 0. \quad (\text{A3})$$

The two eigenvectors can be chosen as

$$\{rM \ rN \ g^2\} e^{rz} \quad \text{and} \quad \{N \ -M \ 0\} e^{rz}, \quad (\text{A4})$$

such that they satisfy equation (A3). Of course, any linear combination of the two eigenvectors given by (A4) is also an eigenvector and can be used to replace any of the eigenvectors (A4). By changing $+r$ to $-r$ in equation (A4) the eigenvectors for the double root $-r$ is obtained.

To obtain the eigenvector corresponding to the single root $+s$ a similar procedure is adopted. Choosing

$$\begin{bmatrix} \phi \\ \psi \\ \chi \end{bmatrix} = \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} e^{sz} \quad (\text{A5})$$

and substituting in equation (6) one obtains

$$\begin{bmatrix} (g^2 - s^2 - M^2) & -MN & Ms \\ -MN & (g^2 - s^2 - N^2) & Ns \\ Ms & Ns & -g^2 \end{bmatrix} \begin{bmatrix} P_1 \\ P_2 \\ P_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}. \quad (\text{A6})$$

The rank of the (3×3) matrix on the left-hand side of equation (A6) is two and thus solution (A5) can yield one eigenvector. It is easily verified that the eigenvector

$$\{M \ N \ s\} e^{sz} \quad (\text{A7})$$

satisfies equation (A6). Changing $+s$ to $-s$ in (A7) the eigenvector for the root $-s$ is obtained.